

Ex.no:6b	DATA VISUALIZATION USING POWER BI

AIM

To visualize and analyze the Vehicle Analysis dataset using Line Chart, Bar Chart, and Pie Chart with Matplotlib to understand attack trends, regional comparisons, and attack distribution.

ALGORITHM

Step 1: Import Pandas and Matplotlib libraries.

Step 2: Load the Vehicle Analysis dataset into a DataFrame.

Step 3: Group the fuel types according to year and calculate the number of vehicle for each year.

Step 4: Plot the yearly vehicle count using a Line Chart to observe the trend of fuel type over time.

Step 5: Group the dataset according to region and calculate the number of vehicles in each region.

Step 6: Plot the regional fuel type counts using a Bar Chart to compare vehicle market across different regions.

Step 7: Select the major regions and calculate their fuel proportions.

Step 8: Plot the regional distribution using a Pie Chart.

Step 9: Add suitable titles, axis labels, legends, and display the visualizations.

CODE

```
import pandas as pd

import matplotlib.pyplot as plt

import kagglehub

import os

# Load Vehicle Dataset from Kaggle

path = kagglehub.dataset_download(

    "nehalbirla/vehicle-dataset-from-cardekho"

)

print("Dataset Path:")
```

```
print(path)

# Find CSV file automatically

csv_files = [f for f in os.listdir(path) if f.endswith(".csv")]

csv_file = csv_files[0]

# Load Dataset

df = pd.read_csv(

    os.path.join(path, csv_file),

    encoding="latin1",

    low_memory=False

)

# Clean column names

df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")

print("\nDataset Loaded Successfully")

# -----

# 1. Line Chart – Vehicles by Year

# -----

yearly_vehicles = df.groupby("year").size()

plt.figure(figsize=(10, 5))

plt.plot(

    yearly_vehicles.index,

    yearly_vehicles.values

)

plt.title("Vehicles by Year")

plt.xlabel("Year")

plt.ylabel("Number of Vehicles")

plt.grid(True)
```

```
plt.show()

# -----

# 2. Bar Chart – Vehicles by Fuel Type

# -----

fuel_vehicles = df["fuel_type"].value_counts()

plt.figure(figsize=(12, 6))

plt.bar(

    fuel_vehicles.index,

    fuel_vehicles.values

)

plt.title("Vehicles by Fuel Type")

plt.xlabel("Fuel Type")

plt.ylabel("Number of Vehicles")

plt.xticks(rotation=45, ha="right")

plt.tight_layout()

plt.show()

# -----

# 3. Pie Chart – Fuel Type Distribution

# -----

top_fuels = fuel_vehicles.head(6)

plt.figure(figsize=(8, 8))

plt.pie(

    top_fuels.values,

    labels=top_fuels.index,

    autopct="%1.1f%%"

)
```

```
plt.title("Top 6 Fuel Types by Number of Vehicles")
```

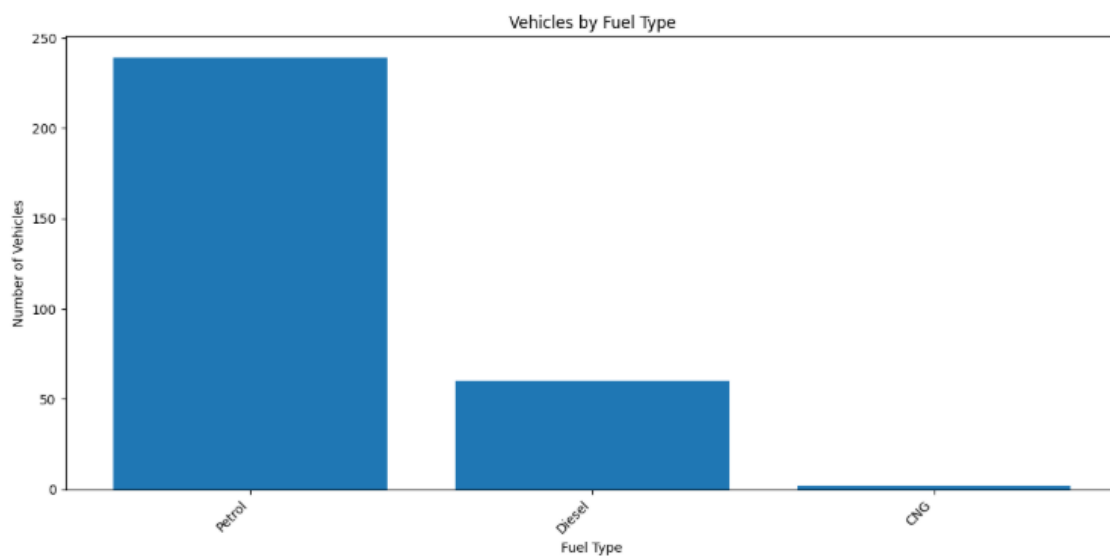
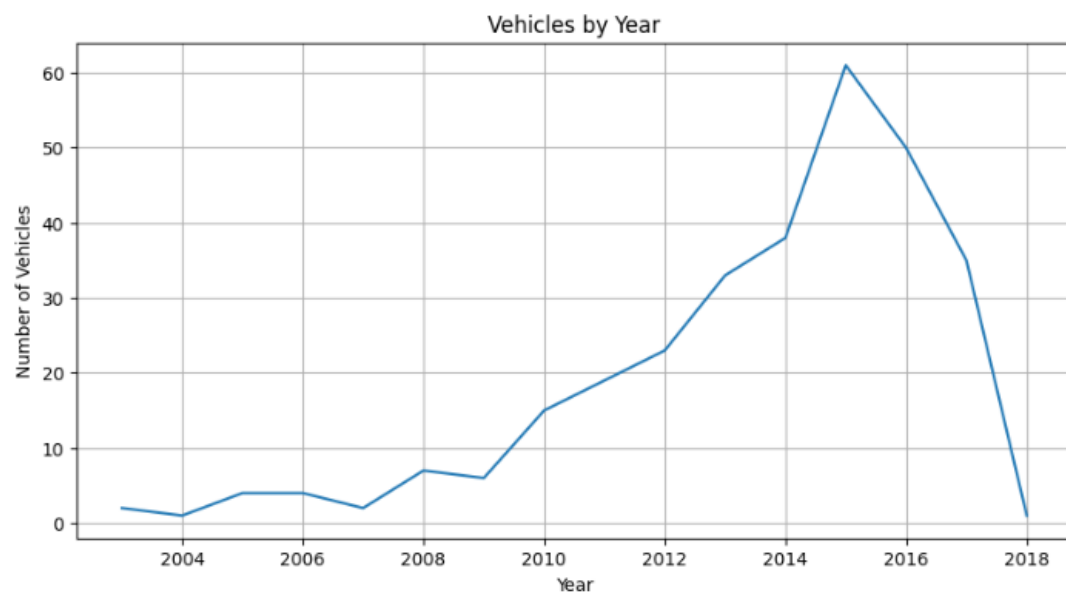
```
plt.show()
```

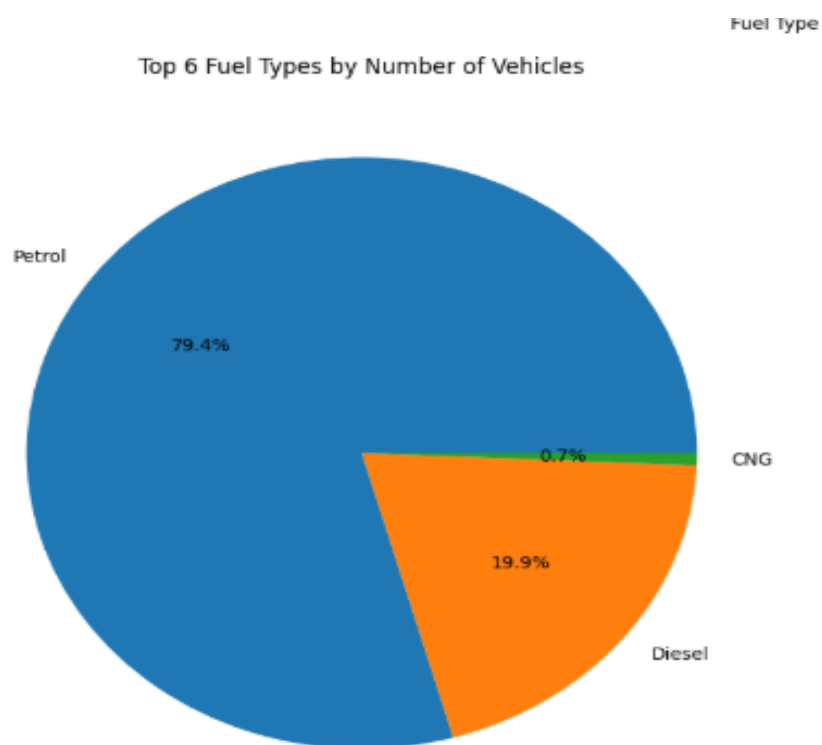
OUTPUT

Dataset Path:

/kaggle/input/vehicle-dataset-from-cardekho

Dataset Loaded Successfully





RESULT